Labour Supply over the Life Cycle

ECON 300 - Labour Economics

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Learning Objectives

- Explain why cross-sectional age comparisons can be misleading.
- Distinguish the effects of **temporary** vs. **permanent** wage changes on labour supply.
- Describe how economists model household choices over paid work, unpaid work, and fertility.
- Outline Canada's three-tier pension system and retirement incentives.
- Use diagrams to show how public pensions affect retirement timing and income.

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Main Questions

- Do men's and women's labour supply patterns evolve similarly with age?
- How do life-cycle planning horizons change labour-supply responses to wages?
- What new complexities arise when modelling each period of life?
- How do labour-saving household technologies reshape time allocation?
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Age–Participation Profiles

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Figure 4.1 — Age–Labour Force Participation Profiles *(Placeholder)*

Figure 4.1 — Plot age-participation profiles for men and women; show cohort shifts.

Insert figure/diagram here later.

Cohort Effect: Definition

- The difficulty of separating the influence of **age** from **year of birth** (vintage) at a point in time.
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Worked Example — What Is a Cohort Effect?

	Observed in 2000	**2005**	**2010**
Older Cohort, born in 1975:			
Age	25	30	35
Participation Rate	75%	80%	85%
Younger Cohort, born 1980:			
Age	20	25	30
Participation Rate	75%	80%	85%

Takeaway: Cross-section comparisons can confound true aging patterns with cohort differences. Tracking cohorts over time isolates life-cycle effects.

Two Modelling Approaches

- **Sequence of static problems:** convenient but ignores forward-looking behaviour.
- **Dynamic framework:** individuals choose consumption and leisure for each of N periods to maximize lifetime utility given an intertemporal budget constraint.

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Assumptions & Setup

- Horizon of N periods; preferences over $(c_t, \ell_t)_{t=1}^{N}$.
- For intuition, start with no nonlabour income and perfect foresight.
- Labour supply each period links through **present-value** budget constraint.

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Lifetime Income Constraint (Two-Period Illustration)

Income each period: $y_t = w_t h_t$, with $h_t = T - \ell_t$.

Let real interest rate be r. Present-value constraint:

$$\mathsf{PV}_0(c) \;=\; \sum_{t=1}^{2} \frac{c_t}{(1+r)^{t-1}} \;=\; \sum_{t=1}^{2} \frac{w_t(T-\ell_t)}{(1+r)^{t-1}} \;=\; \mathsf{PV}_0(y).$$

Equivalently, replacing h_t by ℓ_t: higher wages raise the opportunity cost of leisure across periods.

Figure 4.2 — Dynamic Life-Cycle Wage Changes *(Placeholder)*

Figure 4.2 — Contrast temporary vs. permanent and anticipated vs. unanticipated wage changes; show income vs. substitution effects.

Insert figure/diagram here later.

Temporary vs. Permanent Wage Increases

- **Temporary** increase: strong substitution effect in that period, modest income effect.
- **Permanent** increase: larger lifetime income effect; labour supply response distributed across periods.
- Anticipation matters: planned time shifts differ if change is known in advance.

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Households as Producers and Consumers

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Basic Framework

Utility over produced goods Z_i: $U = f(Z_1, \ldots, Z_n)$, $Z_i = g_i(X_1, \ldots, X_m; h_i)$

- X: market-purchased inputs; h_i: time spent producing Z_i.
- **Leisure** is not purely separate; time is an input into many Z_i.

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Worked Example — Food at Home vs. Precooked

Utility $u = f(\text{nutrition}, \text{rest})$.

- Prices affecting choices:
  - Price of leisure $= W$ (wage).
  - Unit nutrition from raw food: $P_R + Wd$ (goods price + time).
  - Unit nutrition from precooked food: P_C.
- **Comparative statics:**
  - $\downarrow P_C$: more precooked, more leisure; effect on raw food ambiguous.
  - $\downarrow P_R$: more raw food, less leisure; effect on precooked ambiguous.
  - $\uparrow W$: tilt away from time-intensive goods.
  - $\downarrow d$: cheaper home production; leisure/labour supply depends on income vs. substitution.

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Determinants of Fertility

- **Income:** higher income can raise desired children (income effect), but higher potential earnings raise the opportunity cost of time (substitution effect).
- **Cost of children:** foregone earnings, direct expenses.
- **Prices of related goods:** daycare, education, medical care.
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- Implications for private savings, unemployment, labour force size, and national income.

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- Mandatory retirement provisions (compulsory vs. automatic).
- Wealth and earnings:
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  - $\uparrow$ Earnings: income and substitution effects offset; net effect ambiguous.
- Health, job difficulty, and family composition (e.g., dual earners).

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Canada's Pension System: Three Tiers

- **Old Age Security (OAS):** tax-financed demogrant at 65; GIS supplements are means-tested.
- **CPP/QPP:** social insurance funded by compulsory employer/employee contributions; benefits linked to prior earnings.
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Formal Income–Leisure Model with Pension

$$Y_p = B + (1 - p - t)\, W\, (T - \ell),$$

where Y_p: income of a pension receiver; B: pension; W: wage; T: time endowment; ℓ: leisure; p: payroll tax rate; t: implicit tax (clawback).

Figure 4.3 — Budget Constraints under Social Insurance Pensions *(Placeholder)*

Figure 4.3 — Budget lines with pension B and implicit tax/clawback; assume payroll tax $p = 0$ for illustration.

Insert figure/diagram here later.

Employer-Sponsored Pensions: Types

- Flat benefit, earnings-based benefit, and defined-contribution plans.
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Early/Special vs. Postponed Retirement

- Early retirement: often at age ~ 55 with 10+ years of service (less generous due to shorter service).
- Special provisions: ages 60–62 with 20+ years; common in earnings-based plans; extensively subsidized.
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Picking the Perfect Retirement Age *(Placeholder)*

Decision Map — Weigh pension wealth, health, job strain, spouse's work, and implicit taxes.

Insert figure/diagram here later.

Summary

- Life-cycle labour supply shows systematic age patterns; women's profiles have shifted up across cohorts.
- Dynamic models link period-by-period labour supply through a present-value constraint.
- Household production embeds time costs into consumption; technology and wages tilt choices.
- Fertility responds to income, prices of related goods, tastes, and technology, interacting with women's labour supply.
- Canada's pensions (OAS, CPP/QPP, employer/private plans) and plan design shape retirement timing and incomes.

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Thank You!